

--1. Table Creation:

--Write a SQL query to create a table named Books with the following fields:

```
CREATE TABLE IF NOT EXISTS Books (  
    book_id INT Primary Key,  
    title VARCHAR(100),  
    publisher_id INT  
);
```

result of DDL query: 1

--2. Insert Data:

--Write a query to insert one records into the Books table with appropriate values for book_id, title, and publisher_id.

-- (Please use the publisher ID from publisher table because it is configured as foreign key here)

```
INSERT INTO Books (book_id, title, publisher_id) VALUES (1, 'Grapes of Wrath', 1622);
```

result of DDL query: 1

--3. Update the title of a book with book_id = 1 in the Books table to a new title of your choice.

```
UPDATE Books SET title = 'The Grapes of Wrath' WHERE book_id=1;
```

result of DDL query: 1

--4. Write a query to delete the record from the Books table where the book_id is 1.

```
DELETE FROM Books WHERE book_id=1;
```

result of DDL query: 1

--5. Count the number of distinct authors from the author table.

```
SELECT count(DISTINCT au_id) ct_distinct FROM authors;
```

--SQL DML query results:

```
ct_distinct  
23
```

--Should be the same as count(*)

```
SELECT count(*) ct FROM authors;
```

--SQL DML query results:

```
ct  
23
```

--6. Write a query using an inner join with the employee and publishers table. You will then extract the data from both tables.

```
SELECT *
```

```
FROM employee e JOIN publishers p ON e.pub_id=p.pub_id;
```

--SQL DML query results:

```
emp_id, fname, minit, lname, job_id, job_lvl, pub_id, hire_date,  
pub_id, pub_name, city, state, country  
PTC11962M, Philip, T, Cramer, 2, 215, 9952, 1989-11-11 00:00:00, 9952,  
Scootney Books, New York, NY, USA
```

AMD15433F, Ann, M, Devon, 3, 200, 9952, 1991-07-16 00:00:00, 9952,
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9901, GGG&G, München, None, Germany
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9999, Lucerne Publishing, Paris, None, France
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 Algodata Infosystems, Berkeley, CA, USA
 PSP68661F, Paula, S, Parente, 8, 125, 1389, 1994-01-19 00:00:00, 1389,
 Algodata Infosystems, Berkeley, CA, USA
 MAS70474F, Margaret, A, Smith, 9, 78, 1389, 1988-09-29 00:00:00, 1389,
 Algodata Infosystems, Berkeley, CA, USA
 A-C71970F, Aria, , Cruz, 10, 87, 1389, 1991-10-26 00:00:00, 1389,
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 MAP77183M, Miguel, A, Paolino, 11, 112, 1389, 1992-12-07 00:00:00,
 1389, Algodata Infosystems, Berkeley, CA, USA
 Y-L77953M, Yoshi, , Latimer, 12, 32, 1389, 1989-06-11 00:00:00, 1389,
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 CGS88322F, Carine, G, Schmitt, 13, 64, 1389, 1992-07-07 00:00:00,
 1389, Algodata Infosystems, Berkeley, CA, USA
 PSA89086M, Pedro, S, Afonso, 14, 89, 1389, 1990-12-24 00:00:00, 1389,
 Algodata Infosystems, Berkeley, CA, USA
 A-R89858F, Annette, , Roulet, 6, 152, 9999, 1990-02-21 00:00:00,
 9999, Lucerne Publishing, Paris, None, France
 HAN90777M, Helvetius, A, Nagy, 7, 120, 9999, 1993-03-19 00:00:00,
 9999, Lucerne Publishing, Paris, None, France
 M-P91209M, Manuel, , Pereira, 8, 101, 9999, 1989-01-09 00:00:00,
 9999, Lucerne Publishing, Paris, None, France
 KJJ92907F, Karla, J, Jablonski, 9, 170, 9999, 1994-03-11 00:00:00,
 9999, Lucerne Publishing, Paris, None, France
 POK93028M, Pirkko, O, Koskitalo, 10, 80, 9999, 1993-11-29 00:00:00,
 9999, Lucerne Publishing, Paris, None, France
 PCM98509F, Patricia, C, McKenna, 11, 150, 9999, 1989-08-01 00:00:00,
 9999, Lucerne Publishing, Paris, None, France

--7. Write a query using a left join with the employee and publishers table. You will then extract all the data from the employee table.

SELECT *

FROM employee e LEFT JOIN publishers p ON e.pub_id=p.pub_id;

--SQL DML query results:

emp_id, fname, minit, lname, job_id, job_lvl, pub_id, hire_date,

pub_id, pub_name, city, state, country

PTC11962M, Philip, T, Cramer, 2, 215, 9952, 1989-11-11 00:00:00, 9952,
 Scootney Books, New York, NY, USA

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 PCM98509F, Patricia, C, McKenna, 11, 150, 9999, 1989-08-01 00:00:00,
 9999, Lucerne Publishing, Paris, None, France

--8. Write a query using a right join with the employee and publishers table. You will then extract all the data from publisher table.

SELECT *

FROM employee e RIGHT JOIN publishers p ON e.pub_id=p.pub_id;

--SQL DML query results:

emp_id, fname, minit, lname, job_id, job_lvl, pub_id, hire_date,
 pub_id, pub_name, city, state, country
 PTC11962M, Philip, T, Cramer, 2, 215, 9952, 1989-11-11 00:00:00, 9952,
 Scootney Books, New York, NY, USA
 AMD15433F, Ann, M, Devon, 3, 200, 9952, 1991-07-16 00:00:00, 9952,
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 9999, Lucerne Publishing, Paris, None, France
 PCM98509F, Patricia, C, McKenna, 11, 150, 9999, 1989-08-01 00:00:00,
 9999, Lucerne Publishing, Paris, None, France

--9. Write an ALTER TABLE query to add a new column named author (VARCHAR(100)) to the Books table.

ALTER TABLE Books ADD COLUMN author varchar(100);
 result of DDL query: 1

--10. Write a SQL query to drop the Books table from the database.

DROP TABLE Books;
 result of DDL query: 1